ANALYTICAL CUMULATIVE EXAMINATION
January 19, 1991

Consider the titration of 50.00 mL of 0.0100 F Cu^{+2} with 0.0200 F EDTA at a pH of 8.00. The formation constant for the chelation of copper is 6.3×10^{18} and $\alpha_4 = 5.4 \times 10^{-3}$.

1. What is pCu when 10.00 mL of titrant has been added?
2. What is pCu when 25.00 mL of titrant has been added?
3. What is pCu when 30.00 mL of titrant has been added?

Consider the following cell and the given standard reduction potentials.

$Pt/Fe^{+2}(0.0500\ F),\ Fe^{+3}(0.500\ F)//KCl(1.00\ F)/AgCl(sat'd)/\ Ag$

$Fe^{+3} + e \rightarrow Fe^{+2}\ E^o = 0.771\ V$ $\qquad$ $AgCl + e \rightarrow Ag + Cl^-\ E^o = 0.222\ V$

4. What is E^o for the cell?
5. What is E for the cell?
6. Write a balanced equation for the spontaneous, overall cell reaction.
7. What is the equilibrium constant for the reaction?

Consider the titration of 10.00 mL of 0.100 F Sn^{+2} with 0.100 F Ce^{+4} and in 1 F HNO_3 solution where the formal reduction potentials are:

$Ce^{+4} + e \rightarrow Ce^{+3}\ E_f = 1.61\ V$ $\qquad$ $Sn^{+4} + 2e \rightarrow Sn^{+2}\ E_f = 0.15\ V$

Consider also that the indicating electrode is a platinum flag and the reference is a standard hydrogen electrode.

8. What is E for the cell when 10.00 mL of titrant has been added?
9. What is E for the cell when 20.00 mL of titrant has been added?
10. What is E for the cell when 30.00 mL of titrant has been added?